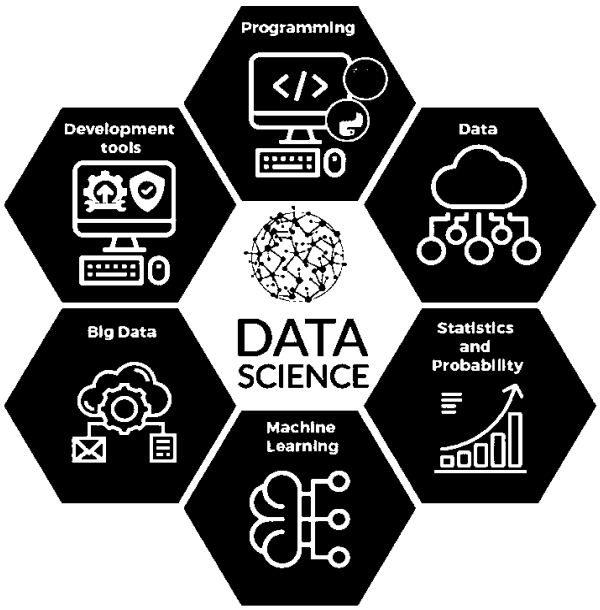

DATA SCIENCE & AI

MIDS PROJECT REPORT

ABSTRACT
This project applies various machine learning models to predict cardiovascular disease, identifying Random Forest as the most accurate and reliable classifier.

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Table of Contents

1. **Project Overview**
2. **Data Collection**
3. **Exploratory Data Analysis**
4. **Realistic Data Collection**
5. **Data Pre-processing**
   - i. **Data Transformation**
   - ii. **Data Cleaning**
     - a) Value Imputation
     - b) Outlier Detection and Removal
   - iii. **Data Reduction**
     - a) Exploratory Data Analysis
     - b) Feature Selection Process
6. **Model Training and Evaluation**
   - i. **Test Train Split and Randomness**
   - a) **Baseline Machine Learning Models**
     - i. Logistic Regression
     - ii. Naïve Bayes
   - b) **Decision Tree Machine Learning Model**
     - i. Decision Tree Classifier
     - ii. Random Forest Classifier
     - iii. AdaBoost Classifier
     - iv. XGBoost Classifier
     - v. LightGBM Classifier
     - vi. CatBoost Classifier
   - c) **Distance and Marginal Based Model**
     - i. K Nearest Neighbors
     - ii. Support Vector Machine
7. **Inference Models**

CARDIO DISEASE PREDICTION

MACHINE LEARNING MODEL TRAINING

SUPERVISED CLASSIFICATION PROBLEM

1. Project Overview

This data science project focuses on **cardiovascular disease prediction**. The objective of the project is to develop a predictive model capable of assessing whether an individual is likely to suffer from cardiovascular disease based on various medical and lifestyle-related factors. The project involves **data collection**, **data pre-processing**, **exploratory data analysis (EDA)**, **model training** and **evaluation** to identify relevant features that significantly contribute to the target prediction.

2. Data Collection

The first dataset used for analysis was obtained from **Kaggle**, containing approximately 7,000 record**s** and the following features:

Age	Gender	Cholesterol	Glucose
Smoke	Alcohol	Active	Low Blood Pressure,
High Blood Pressure	Height	Weight	Target

The target variable was **cardio disease**, labelled as 0 (no disease) and 1 (disease present).

3. EDA for Feature Selection

An initial exploratory data analysis was conducted using **Seaborn regression plots** and **correlation heatmaps** to investigate relationships between independent features and the target variable.

Most features did not demonstrate any meaningful positive or negative correlation with the target variable. Variables such as **Height**, **Weight**, **Glucose**, and **Cholesterol** exhibited weak or no relationship with cardiovascular outcomes. Only **Blood Pressure (low and high)** and **Age** showed a mild positive correlation, suggesting a slight increase in disease probability with increasing age and blood pressure.

Further investigation revealed that **Glucose** and **Cholesterol** were encoded as **categorical values (1, 2, 3)** instead of numerical readings (such as mg/dL). This categorical representation hindered accurate visualization and correlation analysis, as the encoded values did not represent true quantitative differences.

Consequently, the feature selection process based on this dataset produced unreliable results, prompting the need for a more realistic dataset.

4. Realistic Dataset Collection for Cardio Disease Prediction

A more medically relevant dataset was obtained from the **UCI Machine Learning Repository (UCIMLRepo)**. Although smaller in size (approximately **313** records), this dataset contained more realistic and continuous features commonly used in medical diagnosis.

The dataset included the following features:

Feature	***Description***
age	Age of the patient (in years).
sex	Gender of the patient (1 = male, 0 = female).
cp (chest pain type)	Describes the type of chest pain (1–4: typical angina, atypical angina, non-anginal pain, asymptomatic).
trestbps (resting blood pressure)	Resting blood pressure in mm Hg measured upon hospital admission.
chol (serum cholesterol)	Serum cholesterol in mg/dL.
fbs (fasting blood sugar)	Indicates if fasting blood sugar > 120 mg/dL (1 = true, 0 = false).
restecg (resting electrocardiographic results)	Resting ECG results, categorized from normal to abnormal (0–2).
thalach (maximum heart rate achieved)	Maximum heart rate achieved during exercise.
exang (exercise induced angina)	Exercise-induced chest pain (1 = yes, 0 = no).
oldpeak	ST depression induced by exercise relative to rest (ECG measurement).
slope	The slope of the peak exercise ST segment (upsloping, flat, or down sloping).
ca	Number of major vessels (0–3) coloured by fluoroscopy.
thal (thalassemia)	A blood disorder measured as 3 = normal, 6 = fixed defect, 7 = reversible defect.
target	Indicates presence or absence of heart disease.

5. Data Pre-Processing

i. Feature Transformation

The dataset contained **five classes** in the target variable for cardiovascular disease: **0, 1, 2, 3,** and 4. These classes represented varying degrees of heart disease severity.

To simplify the prediction process, a lambda function was applied to the target column to consolidate the classes. The function converted classes 1, 2, 3, and 4 into 1 (disease present) while retaining 0 as 0 (no disease). This transformation streamlined the problem into a binary classification task, allowing for clearer interpretation and improved model performance.

ii. Data Cleaning

An assessment of data completeness revealed only three missing values. To maintain dataset integrity, **value imputation** was performed**.**

a) Value Imputation

Value imputation is a statistical method used to replace missing data points with estimated values derived from existing data. This prevents information loss and ensures consistency across features.
Common imputation methods include:

- **Mean imputation**: Replaces missing values with the column's average (suitable for continuous data).
- **Median imputation**: Uses the median value, reducing sensitivity to outliers.
- **Mode imputation**: Replaces missing values with the most frequent category (suitable for categorical data).

In this dataset, **mode imputation** was applied because the missing entries were found in categorical columns. This approach ensured that imputed values remained consistent with the feature's categorical distribution.

b) Outlier Detection and Removal

Outliers can distort statistical relationships and reduce model accuracy. To ensure data quality, outliers were identified and treated using the **Interquartile Range (IQR) method** and **logarithmic transformation**.

Outlier Detection using the IQR Method

The **Interquartile Range (IQR)** method is a statistical technique used to detect extreme values.
It is calculated as:

$$\boxed{IQR = Q3 - Q1}$$

where **Q1** and **Q3** represent the 25th and 75th percentiles, respectively.

Any value that falls below:

$$\boxed{Q1 - 1.5 \times IQR}$$

or above:

$$\boxed{Q3 + 1.5 \times IQR}$$

is considered an **outlier**.

This method was applied to numeric features such as **cholesterol** and **resting blood pressure**, where several outliers were detected.

Log Transformation

To reduce the impact of extreme values, a **logarithmic transformation** was applied using NumPy's function.

np.log()

This transformation compresses large numerical values and normalizes data distribution, minimizing skewness. As a result, the transformed features become more suitable for modelling, leading to improved prediction stability and accuracy.

iii. Data Reduction

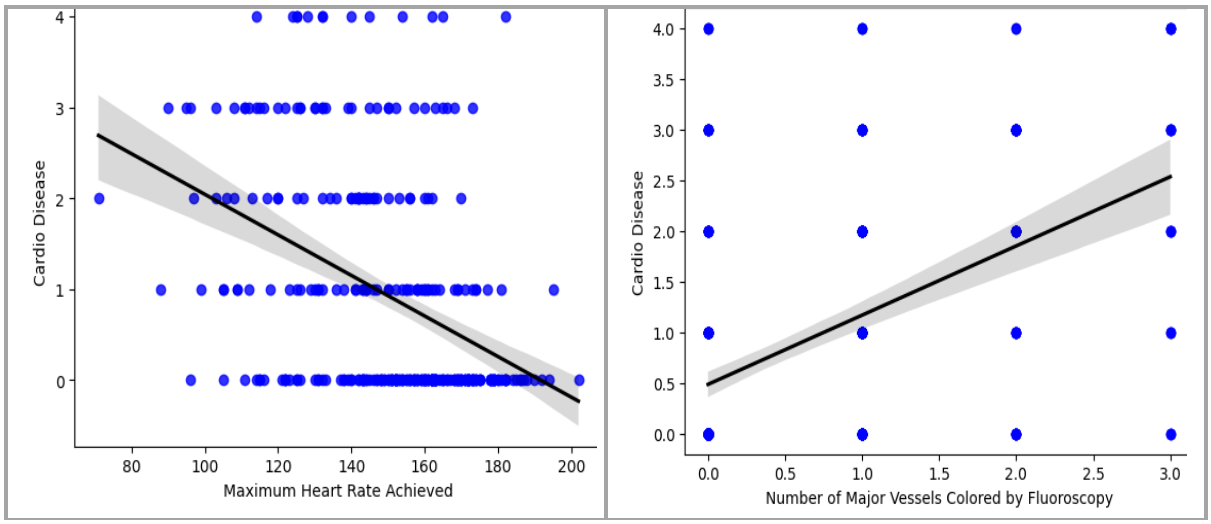
a) Exploratory Data Analysis

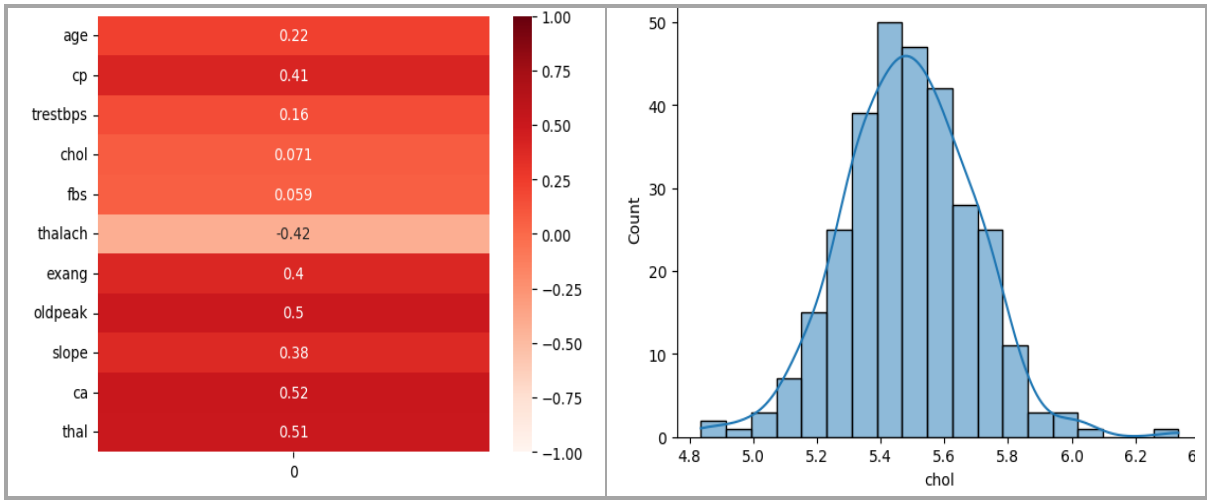
Exploratory Data Analysis (EDA) was performed using **Matplotlib** and **Seaborn** libraries to identify relationships between features and the target.

Regression plots were generated for each independent variable against the target variable to observe their direction and strength of correlation. The regression line in each plot visually indicated whether a feature had a **positive**, **negative**, or **neutral** relationship with cardiovascular disease presence.

The **heatmap** visually represented the correlation coefficients between all variables and the target. Features such as **fbs**, **slope**, and **exang** showed weak correlations, making them less suitable for inclusion in the final model. Conversely, **age**, **cp**, **ca**, and **thalach** were among the most influential features.

Few charts to showcase EDA for best feature selection:





a) Feature Selection Process

Identified relationship of features with the target variable after exploratory data analysis and selected almost all features except the ones that had no significance.

Features	***Relationship***
Age	a **positive relationship,** confirming that risk increases with age.
Chest Pain Type (cp)	**positive correlation**, indicating certain pain types are strong disease indicators.
Resting Blood Pressure (trestbps)	**mild positive relationships**
Cholesterol (chol)	**mild positive relationships**
Fasting Blood Sugar (fbs)	**no significant relationship**, as the regression line remained almost flat.
Maximum Heart Rate Achieved (thalach)	a **negative correlation**, meaning higher maximum heart rate corresponds to lower disease probability
Number of Major Vessels (ca)	**positive correlations** with disease presence.
Oldpeak	displayed a **negative trend**, suggesting that higher ST depression during exercise is associated with heart abnormalities.

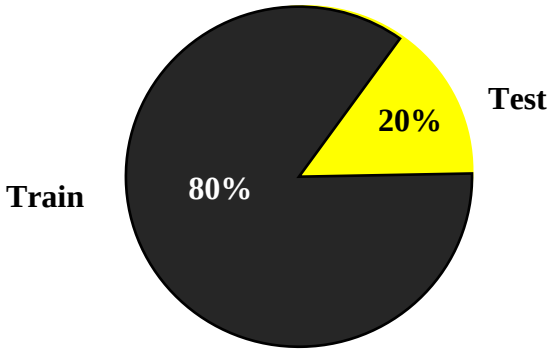
Exercise Induced Angina (exang)	a **positive correlation**, implying higher disease likelihood for individuals experiencing angina during physical exertion.
Thalassemia (thal)	**positive correlations** with disease presence.

6. Model Training and Evaluation

a) Test-Train split and Randomness

The dataset was divided into training and testing subsets to evaluate model performance effectively. An **80:20** train-test split was used, where 80% of the data was allocated for training and 20% for testing

Random state of **42**; means that the random shuffling of data would remain consistent every time the experiment is run**.**



b) Baseline Machine Learning Models

i. Logistic Regression Model

It is widely used algorithms for binary classification problems due to its simplicity, interpretability, and ability to provide clear probabilistic outputs. Regression works by establishing a linear relationship between the input features and the log-odds of the target variable, making it highly suitable for determining whether a person is at risk of cardiovascular disease (class 1) or not (class 0).

Model Performance

Accuracy	0.8688
Confusion Matrix	$\begin{bmatrix} 27 & 6 \\ 2 & 26 \end{bmatrix}$

F1 Score	0.8667

Training and Testing Accuracy

- **Training Accuracy:** 0.8595
- **Testing Accuracy:** 0.8689

Training and testing accuracy demonstrate that the model is **well-generalized**. It is neither **Overfitting:** performing exceptionally well on training data but poorly on unseen data. **Underfitting:** failing to capture the underlying patterns in the data.

ii. Naive Bayes Model

Naive Bayes is widely used in classification problems because of its simplicity, computational efficiency, and strong performance on small to medium-sized datasets. It helps determine how feature distributions contribute to the classification of a person as having or not having a cardiovascular condition.

Model Performance

Accuracy	0.8525
Confusion Matrix	$\begin{bmatrix} 25 & 8 \\ 1 & 27 \end{bmatrix}$
F1 Score	0.8571

Training and Testing Accuracy

- **Training Accuracy:** 0.8302
- **Testing Accuracy:** 0.8571

It shows model has **consistent performance** on both seen and unseen data.

c) Decision Tree Machine Learning Models

i. Decision Tree Classifier

Its primary purpose is to model decisions and their possible outcomes in a tree-like structure of nodes and branches. Each internal node represents a **feature-based decision**, each branch represents an **outcome** of that decision.

Hyperparameter Tuning

To prevent overfitting and improve generalization, several hyperparameters were tuned:

- **max_depth = 10:** Limits the depth of the tree to prevent it from becoming too complex.
- **criterion = 'gini':** Uses the Gini Impurity measure to decide the best feature splits for classification.

- **max_leaf_nodes = 5:** Restricts the maximum number of terminal nodes (leaves) in the tree to simplify the model.
- **min_samples_split = 10:** Ensures that a node is only split if it contains at least 10 samples, reducing the likelihood of creating overly specific splits.

Model Performance

Accuracy	0.8
Confusion Matrix	$\begin{bmatrix} 31 & 2 \\ 8 & 20 \end{bmatrix}$
F1 Score	0.8360

Training and Testing Accuracy

- **Training F1:** 0.78
- **Testing F1:** 0.8

The training and testing accuracies are almost close, indicating that the model is **slightly overfitting.**

ii. Random Forest Classifier

It is based on the principle of "wisdom of the crowd," where multiple weak learners (individual trees) collectively form a strong predictive model. Random Forest reduces overfitting that is often seen in individual decision trees and improves model robustness and generalization.

Hyperparameter Tuning

- **max_depth = None:** Allows each decision tree to grow until all leaves are pure, letting the model automatically determine the ideal depth.
- **n_estimators = 200:** Builds 200 decision trees to ensure stability and minimize variance across predictions.
- **min_samples_leaf = 5:** Sets the minimum number of samples required at a leaf node, preventing the model from learning insignificant noise.
- **min_samples_split = 6:** Ensures a node must have at least six samples before it can be split, promoting smoother and more general splits.
- **random_state = 42:** Provides reproducibility by ensuring consistent results each time the model is trained.

Model Performance

Accuracy	0.8333
Confusion Matrix	$\begin{bmatrix} 31 & 1 \\ 9 & 19 \end{bmatrix}$
F1 Score	0.7961

Training and Testing Accuracy

- **Training F1:** 0.8761
- **Testing F1:** 0.8666

The training and testing accuracies are relatively close, indicating that the model is **well-generalized.**

iii. AdaBoost Classifier

The **AdaBoost (Adaptive Boosting) Classifier** is an ensemble learning algorithm that combines multiple weak learners; typically, shallow decision trees to create a strong predictive model.

Hyperparameter Tuning

- **n_estimators = 100:** The model builds 100 weak learners (decision stumps) sequentially, with each contributing to the final decision based on its performance.
- **learning_rate = 1.0:** Controls how much each weak learner influences the final prediction. A value of 1.0 is the default and provides a balanced learning pace.

Model Performance

Accuracy	0.75
Confusion Matrix	$\begin{bmatrix} 27 & 5 \\ 5 & 18 \end{bmatrix}$
F1 Score	0.705
ROC-AUC Score	0.74

Training and Testing Accuracy

- **Training Accuracy:** 0.9315
- **Testing Accuracy:** 0.7058

The high training accuracy (0.93) compared to testing accuracy (0.70) suggests that the model may exhibit **overfitting**, meaning it performed very well on training data but struggled to generalize to unseen data.

iv. XGBClassifier

XGBoost is known for its strong regularization mechanisms that help prevent overfitting, making it one of the most powerful ensemble methods for structured data tasks such as medical prediction.

Hyperparameter Tuning

- **n_estimators = 200:** Builds 200 boosting rounds (trees) to improve predictive strength.
- **learning_rate = 0.1:** Determines the contribution of each tree, balancing learning speed and model accuracy.
- **max_depth = 4:** Limits the depth of each tree to control complexity and prevent overfitting.
- **subsample = 0.8:** Uses 80% of the training samples per tree, increasing model robustness.

- **colsample_bytree = 0.8:** Randomly selects 80% of the features for each tree to reduce correlation among trees.
- **random_state = 42:** Ensures reproducibility of results.
- **use_label_encoder = False:** Disables deprecated label encoding.
- **eval_metric = 'logloss':** Uses logarithmic loss to evaluate prediction accuracy during training.

Model Performance

Accuracy	0.8166
Confusion Matrix	$\begin{bmatrix} 29 & 3 \\ 8 & 20 \end{bmatrix}$
F1 Score	0.7843

Training and Testing Accuracy

- **Training Accuracy:** 1.0
- **Testing Accuracy:** 0.7843

The perfect training accuracy **(1.0)** compared to a testing accuracy of **0.78** suggests that the model might be **slightly overfitting.**

v. LightGBM Classifier

LightGBM uses a **leaf-wise tree growth strategy**, which splits the tree based on the leaf with the highest loss reduction, resulting in deeper trees and improved accuracy while maintaining computational efficiency.

Hyperparameter Tuning

- **n_estimators = 200:** The model builds 200 boosting iterations to progressively improve prediction accuracy.
- **learning_rate = 0.1:** Controls how much each new tree contributes to the model, balancing learning speed and precision.
- **max_depth = 4:** Restricts the depth of each tree to prevent overfitting and maintain interpretability.

Model Performance

Accuracy	0.8360
Confusion Matrix	$\begin{bmatrix} 26 & 7 \\ 3 & 25 \end{bmatrix}$
F1 Score	0.8333
ROC-AUC Score	0.84

Training and Testing Accuracy

- **Training Accuracy:** 1.0
- **Testing Accuracy:** 0.8333

The **training accuracy of 1.0** suggests slight **overfitting**, the model still maintained **high generalization** on the testing data.

vi. CatBoost Classifier

Categorical Boosting uses ordered boosting and oblivious decision trees to prevent overfitting and reduce prediction bias, making it particularly effective for structured datasets like medical diagnostics.

Hyperparameter Tuning

- **iterations = 200:** Specifies the number of boosting rounds used to iteratively minimize error and improve accuracy.
- **learning_rate = 0.1:** Controls the contribution of each new tree to the overall model, balancing training speed and accuracy.
- **depth = 4:** Defines the maximum depth of each tree, limiting model complexity and preventing overfitting.
- **random_seed = 42:** Ensures reproducible results.
- **verbose = 0:** Suppresses detailed training logs for cleaner output.

Model Performance

Accuracy	0.8852
Confusion Matrix	$\begin{bmatrix} 27 & 6 \\ 1 & 27 \end{bmatrix}$
F1 Score	0.8852

Training and Testing Accuracy

- **Training Accuracy:** 1.0
- **Testing Accuracy:** 0.8852

While the training accuracy of **1.0** suggests that the model fits the training data perfectly, the testing accuracy remained high, showing **excellent generalization capability** with minimal overfitting.

d) Distance and Margin Based Machine Learning Models

i. K-Nearest Neighbors

The K-Nearest Neighbors (KNN) Classifier is a simple, non-parametric, and instance-based learning algorithm that classifies a data point based on the majority class of its nearest neighbors in the feature space.

Hyperparameter Tuning

- **n_neighbors = 5:** Considers the five closest data points (neighbors) when determining the class of a new instance.

- **p = 2:** Uses the **Euclidean distance** metric, as p=2p = 2p=2 corresponds to the standard distance formula.
- **metric = 'minkowski':** A generalized distance metric that allows flexibility in distance measurement; for p=2p = 2p=2, it becomes equivalent to Euclidean distance.

Model Performance

Accuracy	*0.*50
Confusion Matrix	$\begin{bmatrix} 17 & 12 \\ 18 & 13 \end{bmatrix}$
F1 Score	*0.46*

KNN serves as a good baseline for comparison due to its simplicity, it was not suitable for this cardiovascular disease prediction task**.**

ii. Support Vector Classifier (SVC)

Support Vector Machines (SVM) aims to find the optimal hyperplane that best separates the classes in the feature space. By using different kernel functions, SVC can handle both linear and non-linear classification problems effectively.

Hyperparameter Tuning

- **kernel = 'rbf':** The **Radial Basis Function (RBF)** kernel allows the model to handle non-linear relationships by mapping data into a higher-dimensional space where a linear separation is possible.
- **C = 1.0:** The **regularization parameter** that controls the trade-off between achieving a smooth decision boundary and correctly classifying training points. A smaller C value makes the margin wider (more general), while a larger C value tries to classify every training example correctly, potentially leading to overfitting.

Model Performance

Accuracy	*0.*50
Confusion Matrix	$\begin{bmatrix} 26 & 3 \\ 27 & 4 \end{bmatrix}$
F1 Score	*0.2105*

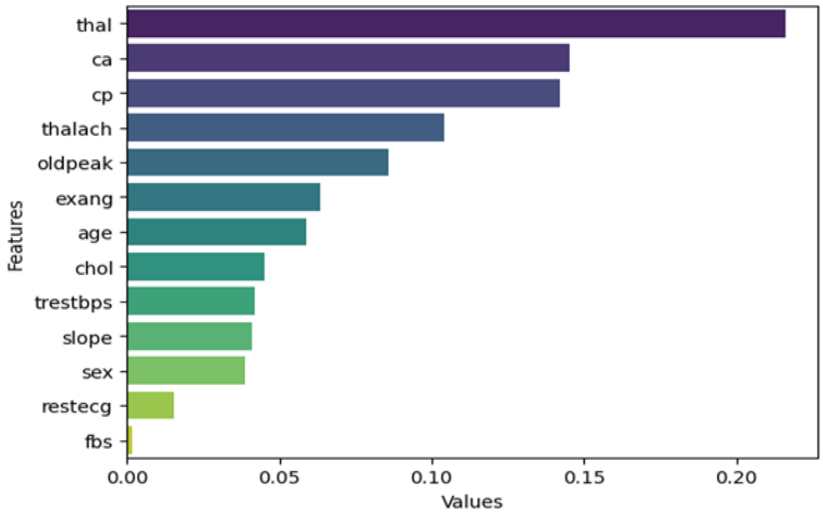
It was not suitable for this cardiovascular disease prediction task due to its low accuracy**.**

7. Inference Model and Pipeline

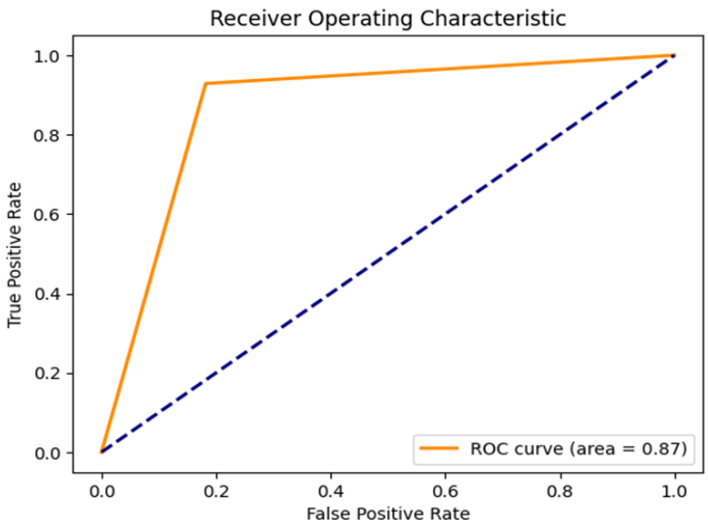
Among all the machine learning models tested, the **Random Forest Classifier** performed the best in predicting cardiovascular disease with **F1 score** of 0.86 and **accuracy** of 0.88. Because of its solid result, the final code will be used as a pipeline in web application made in Flask.

- **Training F1:** 0.8761
- **Testing F1:** 0.8666

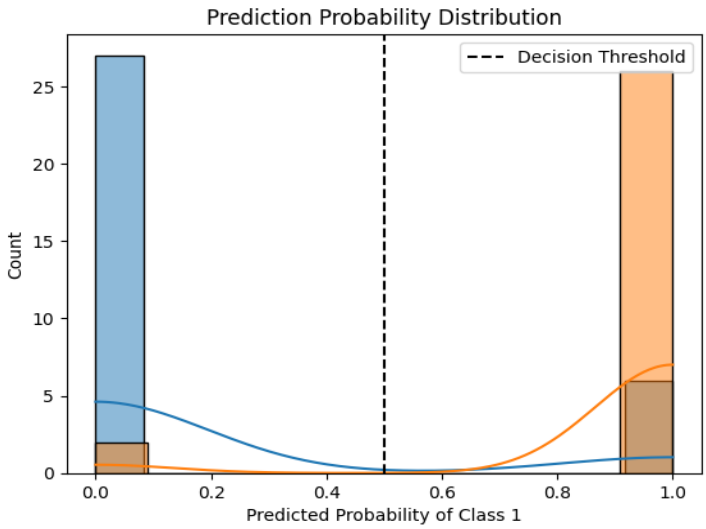
Feature Importance Chart in Random Forest Classifier indicates that gender, fasting blood sugar and resting blood pressure do not play much part in predictions while maximum heart rate, number of major vessels, chest paint and thalassemia type play the most part in predictions.



Feature Importance in Random Forest Classifier



ROC-AUC Curve



Predicted Probability Histogram